

Supplementary Material: A formal knowledge base for metagenomics and geochemistry of the Rub' al Khali desert

1 Supplementary Tables

Table 1: Site-specific environmental measurements across all expeditions. Altitude data was not available.

Expedition	Site	Date	Time	Coordinates	Temp (°C)	Press. (mbar)	Humidity (%)
Trip1 2023	1	17/03/2023	07:58	19.1498 N, 45.1665 E	20.7	928	62
Trip1 2023	2	17/03/2023	08:21	19.2035 N, 45.2448 E	21.3	928	60
Trip1 2023	3	17/03/2023	08:50	19.3011 N, 45.3844 E	22.6	928	55
Trip1 2023	4	17/03/2023	09:20	19.3796 N, 45.5518 E	24	928	51
Trip1 2023	5	17/03/2023	10:29	19.4072 N, 45.6595 E	28.8	928	40
Trip1 2023	6	17/03/2023	11:07	19.4057 N, 45.8161 E	29	928	42
Trip1 2023	7	17/03/2023	11:35	19.4070 N, 45.9539 E	28.7	928	49
Trip1 2023	8	17/03/2023	12:05	19.3932 N, 46.1107 E	30.6	928	41
Trip1 2023	9	17/03/2023	12:44	19.4112 N, 46.3027 E	35.9	928	36
Trip1 2023	10	17/03/2023	13:12	19.4205 N, 46.4608 E	32.5	928	39
Trip1 2023	11	17/03/2023	13:39	19.4444 N, 46.6356 E	31.1	930	37
Trip1 2023	12	17/03/2023	14:04	19.4779 N, 46.8344 E	34.6	932	30
Trip1 2023	13	17/03/2023	15:42	19.5293 N, 47.0533 E	28.9	936	29
Trip1 2023	14	17/03/2023	16:09	19.5813 N, 47.2920 E	30.2	938	37
Trip1 2023	15	17/03/2023	16:33	19.6137 N, 47.4759 E	31.6	946	33
Trip1 2023	16	17/03/2023	16:55	19.6445 N, 47.6261 E	20.7	948	33
Trip1 2023	17	17/03/2023	17:23	19.6765 N, 47.8273 E	20.5	952	32
Trip1 2023	18	18/03/2023	06:49	19.7325 N, 48.0469 E	20	954	50
Trip1 2023	19	18/03/2023	07:15	19.8123 N, 48.2405 E	22.9	960	51
Trip1 2023	20	18/03/2023	07:55	19.8693 N, 48.4246 E	24.3	964	39
Trip1 2023	21	18/03/2023	08:20	19.9011 N, 48.4865 E	29.4	962	30
Trip1 2023	22	18/03/2023	08:45	19.9787 N, 48.6394 E	29	-	25
Trip1 2023	23	18/03/2023	09:25	20.0493 N, 48.7733 E	26.8	966	25
Trip1 2023	24	18/03/2023	09:48	20.0927 N, 48.8994 E	29.5	966	25
Trip1 2023	25	18/03/2023	10:15	20.1003 N, 48.9542 E	27.8	968	27
Trip1 2023	26	18/03/2023	10:40	20.1488 N, 49.0787 E	31.7	968	23
Trip1 2023	27	18/03/2023	11:18	20.2039 N, 49.1960 E	31.5	970	23
Trip1 2023	28	18/03/2023	11:47	20.2438 N, 49.3709 E	31.9	970	24
Trip1 2023	29	18/03/2023	12:15	20.2598 N, 49.5675 E	31.7	968	25
Trip1 2023	30	18/03/2023	14:39	20.2693 N, 49.6837 E	34.1	964	22
Trip1 2023	31	18/03/2023	15:11	20.2946 N, 49.8914 E	33.8	964	22

Expedition	Site	Date	Time	Coordinates	Temp (°C)	Press. (mbar)
Trip1 2023	32	18/03/2023	15:44	20.3107 N, 50.0930 E	34	968
Trip1 2023	33	18/03/2023	16:15	20.3175 N, 50.3031 E	34.8	968
Trip1 2023	34	18/03/2023	16:46	20.3444 N, 50.5154 E	34.1	970
Trip1 2023	35	18/03/2023	17:24	20.3718 N, 50.7300 E	33.7	976
Trip1 2023	36	19/03/2023	09:40	20.4232 N, 51.1448 E	29.8	982
Trip1 2023	37	19/03/2023	10:10	20.4542 N, 51.3449 E	33.5	984
Trip1 2023	38	19/03/2023	10:30	20.4610 N, 51.3702 E	31.9	984
Trip1 2023	39	19/03/2023	11:05	20.5102 N, 51.5974 E	36.4	984
Trip1 2023	40	19/03/2023	11:40	20.5472 N, 51.8256 E	30.5	980
Trip1 2023	41	19/03/2023	12:16	20.5398 N, 51.9461 E	35.4	980
Trip1 2023	42	19/03/2023	12:41	20.5552 N, 52.1240 E	35.4	984
Trip1 2023	43	19/03/2023	13:05	20.5676 N, 52.2241 E	35.2	984
Trip1 2023	44	19/03/2023	13:40	20.5678 N, 52.4104 E	38.4	984
Trip1 2023	45	19/03/2023	13:59	20.5635 N, 52.4678 E	37.1	986
Trip1 2023	46	19/03/2023	14:29	20.5537 N, 52.6692 E	35.7	987
Trip1 2023	47	19/03/2023	15:00	20.5418 N, 52.8688 E	37.3	986
Trip1 2023	48	19/03/2023	17:00	20.5686 N, 53.0940 E	35.3	984
Trip1 2023	49	19/03/2023	17:30	20.6447 N, 53.2879 E	35	986
Trip1 2023	50	20/03/2023	07:14	20.7139 N, 53.4123 E	29.8	988
Trip1 2023	51	20/03/2023	07:37	20.7987 N, 53.4870 E	30.2	992
Trip1 2023	52	20/03/2023	08:02	20.8515 N, 53.7579 E	31.1	992
Trip1 2023	53	20/03/2023	09:03	20.8515 N, 53.7579 E	34.8	992
Trip1 2023	54	20/03/2023	10:03	20.9202 N, 53.9622 E	32.1	992
Trip1 2023	55	20/03/2023	10:24	20.9023 N, 54.0024 E	33.2	990
Trip1 2023	56	20/03/2023	11:05	20.8216 N, 54.2239 E	30	988
Trip1 2023	57	20/03/2023	11:28	20.7350 N, 54.4176 E	31.3	988
Trip1 2023	58	20/03/2023	11:50	20.6654 N, 54.5933 E	33.4	986
Trip1 2023	59	20/03/2023	12:04	20.6563 N, 54.6359 E	37.1	986
Trip1 2023	60	20/03/2023	12:49	20.7064 N, 54.7321 E	37.1	986
Trip1 2023	water well	20/03/2023	13:16	20.7159 N, 54.7997 E	35.6	984
Trip1 2023	road	20/03/2023	13:53	20.8258 N, 54.9889 E	35.1	982
Trip1 2023	road	20/03/2023	14:20	20.9341 N, 55.1261 E	36	982
Trip1 2023	camground	20/03/2023	14:45	21.0311 N, 55.1598 E	37	984
Trip2 2023	1	14/07/2023	06:10	19.1498 N, 45.1665 E	-	34.5
Trip2 2023	2	14/07/2023	07:12	19.2035 N, 45.2448 E	-	36.7
Trip2 2023	3	14/07/2023	07:16	19.3011 N, 45.3844 E	-	37.0
Trip2 2023	4	14/07/2023	07:36	19.3796 N, 45.5518 E	-	38.6
Trip2 2023	5	14/07/2023	07:43	19.4072 N, 45.6595 E	-	39.2
Trip2 2023	6	14/07/2023	08:58	19.4057 N, 45.8161 E	-	41.2
Trip2 2023	7	14/07/2023	09:20	19.4070 N, 45.9539 E	-	41.3
Trip2 2023	8	14/07/2023	10:20	19.3932 N, 46.1107 E	-	41.9
Trip3 2024	1	17/02/2024	06:54	19.1498 N, 45.1665 E	21.9	898
Trip3 2024	2	17/02/2024	07:21	19.2035 N, 45.2448 E	26.6	904
Trip3 2024	3	17/02/2024	08:05	19.3011 N, 45.3844 E	28.3	914
Trip3 2024	4	17/02/2024	08:40	19.3796 N, 45.5518 E	30.2	922
Trip3 2024	5	17/02/2024	09:51	19.4072 N, 45.6595 E	35.2	926
Trip3 2024	6	17/02/2024	10:45	19.4057 N, 45.8161 E	37.1	942
Trip3 2024	7	17/02/2024	11:28	19.4070 N, 45.9539 E	37	926
Trip3 2024	8	17/02/2024	11:50	19.3932 N, 46.1107 E	39	922
Trip3 2024	9	17/02/2024	12:31	19.4112 N, 46.3027 E	36.4	924
Trip3 2024	10	17/02/2024	13:22	19.4205 N, 46.4608 E	37.1	926
Trip3 2024	11	17/02/2024	13:46	19.4444 N, 46.6356 E	41.2	930
Trip3 2024	12	17/02/2024	14:04	19.4779 N, 46.8344 E	39.9	932
Trip3 2024	13	17/02/2024	15:20	19.5293 N, 47.0533 E	40	940

Expedition	Site	Date	Time	Coordinates	Temp (°C)	Press. (mbar)	Hu
Trip3 2024	14	17/02/2024	15:50	19.5813 N, 47.2920 E	41.2	940	14
Trip3 2024	15	17/02/2024	16:13	19.6137 N, 47.4759 E	38.8	944	13
Trip3 2024	16	17/02/2024	16:34	19.6445 N, 47.6261 E	39.4	948	11
Trip3 2024	17	18/02/2024	08:42	19.6765 N, 47.8273 E	23	956	27
Trip3 2024	18	18/02/2024	09:54	19.7325 N, 48.0469 E	29.2	960	18
Trip3 2024	19	18/02/2024	10:25	19.8123 N, 48.2405 E	27.4	962	21
Trip3 2024	20	18/02/2024	11:05	19.8693 N, 48.4246 E	32.9	-	17
Trip3 2024	21	18/02/2024	12:08	19.9011 N, 48.4865 E	31.3	964	31
Trip3 2024	22	18/02/2024	12:52	19.9787 N, 48.6394 E	31.3	968	21
Trip3 2024	23	18/02/2024	13:35	20.0493 N, 48.7733 E	24.4	966	30
Trip3 2024	24	18/02/2024	14:18	20.0927 N, 48.8994 E	25.5	966	31
Trip3 2024	25	18/02/2024	14:38	20.1003 N, 48.9542 E	31	966	24
Trip3 2024	26	18/02/2024	15:12	20.1488 N, 49.0787 E	31.9	966	25
Trip3 2024	27	18/02/2024	15:45	20.2039 N, 49.1960 E	32.4	968	23
Trip3 2024	28	18/02/2024	16:24	20.2438 N, 49.3709 E	27.4	968	27
Trip3 2024	29	18/02/2024	16:56	20.2598 N, 49.5675 E	27.3	968	27
Trip3 2024	30	19/02/2024	08:10	20.2693 N, 49.6837 E	36.8	976	20
Trip3 2024	31	19/02/2024	08:43	20.2946 N, 49.8914 E	23.1	978	29
Trip3 2024	32	19/02/2024	09:20	20.3107 N, 50.0930 E	26.7	982	23
Trip3 2024	33	19/02/2024	10:07	20.3175 N, 50.3031 E	26.6	984	20
Trip3 2024	34	19/02/2024	10:52	20.3444 N, 50.5154 E	29.9	982	17
Trip3 2024	35	19/02/2024	11:30	20.3718 N, 50.7300 E	28.7	984	17
Trip3 2024	36	19/02/2024	13:58	20.4232 N, 51.1448 E	28.2	982	16
Trip3 2024	37	19/02/2024	14:52	20.4542 N, 51.3449 E	29.1	984	15
Trip3 2024	38	19/02/2024	15:01	20.4610 N, 51.3702 E	29.3	984	15
Trip3 2024	39	19/02/2024	15:56	20.5102 N, 51.5974 E	28.8	984	16
Trip3 2024	40	19/02/2024	16:42	20.5472 N, 51.8256 E	28.2	986	17
Trip3 2024	41	19/02/2024	17:14	20.5398 N, 51.9461 E	29.4	988	16
Trip3 2024	42	20/02/2024	08:15	20.5552 N, 52.1240 E	24.6	990	34
Trip3 2024	43	20/02/2024	08:38	20.5676 N, 52.2241 E	27	992	27
Trip3 2024	44	20/02/2024	09:22	20.5678 N, 52.4104 E	29.1	994	22
Trip3 2024	45	20/02/2024	09:44	20.5635 N, 52.4678 E	32.4	996	18
Trip3 2024	46	20/02/2024	10:44	20.5537 N, 52.6692 E	32.3	994	17
Trip3 2024	47	20/02/2024	11:24	20.5418 N, 52.8688 E	33.2	994	17
Trip3 2024	48	20/02/2024	12:54	20.5686 N, 53.0940 E	49.4	994	9.5
Trip3 2024	49	20/02/2024	14:21	20.6447 N, 53.2879 E	34	994	13
Trip3 2024	50	20/02/2024	14:51	20.7139 N, 53.4123 E	31.5	996	21
Trip3 2024	51	20/02/2024	16:11	20.7987 N, 53.4870 E	35.9	994	16
Trip3 2024	52	20/02/2024	06:57	20.8515 N, 53.7579 E	34.3	994	17
Trip3 2024	53	21/02/2024	06:06	20.8515 N, 53.7579 E	21.9	990	35
Trip3 2024	54	21/02/2024	15:58	20.9202 N, 53.9622 E	32.2	998	23
Trip3 2024	55	21/02/2024	15:41	20.9023 N, 54.0024 E	31.3	998	14
Trip3 2024	56	21/02/2024	15:10	20.8216 N, 54.2239 E	31.6	998	13
Trip3 2024	57	21/02/2024	14:48	20.7350 N, 54.4176 E	34.2	996	16
Trip3 2024	58	21/02/2024	14:29	20.6654 N, 54.5933 E	35.6	992	16
Trip3 2024	59	21/02/2024	14:15	20.6563 N, 54.6359 E	36.8	992	14
Trip3 2024	60	21/02/2024	13:20	20.7064 N, 54.7321 E	37.1	992	14
Trip3 2024	oil spill	17/02/2024	10:25	19.4054 N, 45.7271 E	35.1	-	20
Trip3 2024	water well	17/02/2024	14:23	19.5014 N, 46.9015 E	39.2	-	15
Trip3 2024	Camell	17/02/2024	16:49	19.6540 N, 47.6883 E	37.6	-	25
Trip3 2024	plant gradient	18/02/2024	09:25	19.7092 N, 47.9593 E	25	958	24
Trip3 2024	hot spring	19/02/2024	12:25	20.3972 N, 50.9655 E	32.1	984	14
Trip3 2024	12.5+A	17/03/2024	14:22	19.5015 N, 46.9014 E	33	934	30
Trip3 2024	19.5	18/03/2024	07:20	19.8193 N, 48.2402 E	21.8	960	61

Expedition	Site	Date	Time	Coordinates	Temp (°C)	Press. (mbar)
Trip3 2024	27+	18/03/2024	11:21	20.2039 N, 49.1960 E	31.5	970
Trip3 2024	31urine	18/03/2024	15:14	20.2946 N, 49.8914 E	33.8	964
Trip3 2024	32+	18/03/2024	15:44	20.3107 N, 50.0930 E	34	968
Trip3 2024	41+	19/03/2024	12:16	20.5398 N, 51.9461 E	35.4	980
Trip3 2024	47+	19/03/2024	15:00	20.5418 N, 52.8688 E	37.3	986
Trip3 2024	47+grass	19/03/2024	15:00	20.5418 N, 52.8688 E	37.3	986
Trip3 2024	48+	19/03/2024	17:00	20.5686 N, 53.0940 E	35.3	984
Trip3 2024	48+	19/03/2024	17:00	20.5686 N, 53.0940 E	35.3	984
Trip3 2024	52AD	20/03/2024	08:02	20.8515 N, 53.7579 E	31.1	992
Trip3 2024	54F	20/03/2024	10:16	20.9202 N, 53.9622 E	34.8	990
Trip3 2024	35 + WaterWellRestStop	18/03/2024	18:48	20.3972 N, 50.9657 E	-	976
Trip3 2024	35 + hot spring filter1	19/03/2024	08:03	20.3972 N, 50.9657 E	28.3	980
Trip3 2024	35 + hot spring filter 2	19/03/2024	08:11	20.3972 N, 50.9657 E	28.3	980
Trip4 2024	1	29/08/2024	06:04	19.1498 N, 45.1665 E	30.83	-
Trip4 2024	2	29/08/2024	06:24	19.2035 N, 45.2448 E	31.5	-
Trip4 2024	3	29/08/2024	06:53	19.3011 N, 45.3844 E	33.61	-
Trip4 2024	4	29/08/2024	07:15	19.3796 N, 45.5518 E	35.11	-
Trip4 2024	5	29/08/2024	07:29	19.4072 N, 45.6595 E	36.67	-
Trip4 2024	6	29/08/2024	08:07	19.4057 N, 45.8161 E	37.06	-
Trip4 2024	7	29/08/2024	08:32	19.4070 N, 45.9539 E	37.56	-
Trip4 2024	8	29/08/2024	08:57	19.3932 N, 46.1107 E	37.39	-
Trip4 2024	9	29/08/2024	09:28	19.4112 N, 46.3027 E	39	-
Trip4 2024	10	29/08/2024	09:50	19.4205 N, 46.4608 E	39.56	-
Trip4 2024	11	29/08/2024	10:14	19.4444 N, 46.6356 E	42	-
Trip4 2024	12	29/08/2024	10:35	19.4779 N, 46.8344 E	42.94	-
Trip4 2024	13	29/08/2024	11:30	19.5293 N, 47.0533 E	41.94	-
Trip4 2024	14	29/08/2024	11:50	19.5813 N, 47.2920 E	42.56	-
Trip4 2024	15	29/08/2024	12:06	19.6137 N, 47.4759 E	44.11	-
Trip4 2024	16	29/08/2024	12:21	19.6445 N, 47.6261 E	43.5	-
Trip4 2024	17	29/08/2024	12:49	19.6765 N, 47.8273 E	43.11	-
Trip4 2024	18	29/08/2024	13:24	19.7325 N, 48.0469 E	43.83	-
Trip4 2024	19	29/08/2024	13:52	19.8123 N, 48.2405 E	42.83	-
Trip4 2024	20	29/08/2024	13:57	19.8693 N, 48.4246 E	41.94	-
Trip4 2024	21	29/08/2024	14:32	19.9011 N, 48.4865 E	41.72	-
Trip4 2024	22	29/08/2024	14:50	19.9787 N, 48.6394 E	43.33	-
Trip4 2024	23	29/08/2024	15:21	20.0493 N, 48.7733 E	44.28	-
Trip4 2024	24	29/08/2024	15:55	20.0927 N, 48.8994 E	43.17	-
Trip4 2024	25	29/08/2024	16:45	20.1003 N, 48.9542 E	43.5	-
Trip4 2024	26	29/08/2024	17:02	20.1488 N, 49.0787 E	42.83	-
Trip4 2024	27	29/08/2024	17:28	20.2039 N, 49.1960 E	41.83	-
Trip4 2024	28	29/08/2024	17:56	20.2438 N, 49.3709 E	40.83	-
Trip4 2024	29	29/08/2024	18:21	20.2598 N, 49.5675 E	40.78	-
Trip4 2024	30	30/08/2024	05:51	20.2693 N, 49.6837 E	36.5	-
Trip4 2024	31	30/08/2024	06:08	20.2946 N, 49.8914 E	35.89	-
Trip4 2024	32	30/08/2024	06:30	20.3107 N, 50.0930 E	35.72	-
Trip4 2024	33	30/08/2024	06:50	20.3175 N, 50.3031 E	35.89	-
Trip4 2024	34	30/08/2024	07:11	20.3444 N, 50.5154 E	36.39	-
Trip4 2024	35	30/08/2024	07:25	20.3718 N, 50.7300 E	38.5	-
Trip4 2024	36	30/08/2024	08:01	20.4232 N, 51.1448 E	39.28	-
Trip4 2024	37	30/08/2024	08:47	20.4542 N, 51.3449 E	39.44	-
Trip4 2024	38	30/08/2024	09:14	20.4610 N, 51.3702 E	41.06	-
Trip4 2024	39	30/08/2024	09:20	20.5102 N, 51.5974 E	42.44	-
Trip4 2024	40	30/08/2024	09:49	20.5472 N, 51.8256 E	42.06	-
Trip4 2024	41	30/08/2024	10:25	20.5398 N, 51.9461 E	43.17	-

Expedition	Site	Date	Time	Coordinates	Temp (°C)	Press. (mbar)	Hu
Trip4 2024	42	30/08/2024	10:50	20.5552 N, 52.1240 E	43.67	-	16
Trip4 2024	43	30/08/2024	11:19	20.5676 N, 52.2241 E	42.78	-	17
Trip4 2024	44	30/08/2024	11:40	20.5678 N, 52.4104 E	41.94	-	18
Trip4 2024	45	30/08/2024	12:12	20.5635 N, 52.4678 E	40.94	-	17
Trip4 2024	46	30/08/2024	12:22	20.5537 N, 52.6692 E	44	-	15
Trip4 2024	47	30/08/2024	13:16	20.5418 N, 52.8688 E	42.94	-	15
Trip4 2024	48	30/08/2024	13:57	20.5686 N, 53.0940 E	42.5	-	18
Trip4 2024	49	30/08/2024	14:28	20.6447 N, 53.2879 E	44	-	14
Trip4 2024	50	31/08/2024	06:08	20.7139 N, 53.4123 E	35.61	-	25
Trip4 2024	51	31/08/2024	06:34	20.7987 N, 53.4870 E	36.11	-	25
Trip4 2024	52	31/08/2024	06:54	20.8278 N, 53.5784 E	36.28	-	25
Trip4 2024	53	31/08/2024	09:00	20.8515 N, 53.7579 E	42.61	-	17
Trip4 2024	54	31/08/2024	09:34	20.9202 N, 53.9622 E	42.5	-	17
Trip4 2024	55	31/08/2024	09:48	20.9023 N, 54.0024 E	40.72	-	20
Trip4 2024	56	31/08/2024	10:16	20.8216 N, 54.2239 E	40.44	-	21
Trip4 2024	57	31/08/2024	10:40	20.7350 N, 54.4176 E	42.78	-	18
Trip4 2024	58	31/08/2024	10:58	20.6654 N, 54.5933 E	43.5	-	15
Trip4 2024	59	31/08/2024	11:06	20.6563 N, 54.6359 E	43	-	14
Trip4 2024	60	31/08/2024	11:28	20.7064 N, 54.7321 E	42.89	-	18
Trip5 2025	1	12/10/2025	07:32	19.1498 N, 45.1665 E	26.9	894	24
Trip5 2025	2	12/10/2025	08:05	19.2035 N, 45.2448 E	27.8	902	23
Trip5 2025	3	12/10/2025	08:38	19.3011 N, 45.3844 E	29	912	23
Trip5 2025	4	12/10/2025	09:20	19.3796 N, 45.5518 E	32.4	922	18
Trip5 2025	5	12/10/2025	10:00	19.4072 N, 45.6595 E	33.4	924	19
Trip5 2025	6	12/10/2025	10:50	19.4057 N, 45.8161 E	36.7	916	14
Trip5 2025	7	12/10/2025	11:18	19.4070 N, 45.9539 E	35	918	13
Trip5 2025	8	12/10/2025	11:55	19.3932 N, 46.1107 E	34	918	12
Trip5 2025	9	12/10/2025	12:33	19.4112 N, 46.3027 E	35.5	918	12
Trip5 2025	10	12/10/2025	12:58	19.4205 N, 46.4608 E	35.6	920	10
Trip5 2025	11	12/10/2025	13:30	19.4444 N, 46.6356 E	35.6	926	10
Trip5 2025	12	12/10/2025	13:57	19.4779 N, 46.8344 E	36.3	928	10
Trip5 2025	13	12/10/2025	15:25	19.5293 N, 47.0533 E	36.5	934	10
Trip5 2025	14	12/10/2025	15:51	19.5813 N, 47.2920 E	35	936	11
Trip5 2025	15	12/10/2025	16:22	19.6137 N, 47.4759 E	34.7	940	11
Trip5 2025	16	12/10/2025	16:42	19.6445 N, 47.6261 E	33.9	942	13
Trip5 2025	17	13/10/2025	07:05	19.6765 N, 47.8273 E	26.4	948	32
Trip5 2025	18	13/10/2025	07:42	19.7325 N, 48.0469 E	26.6	950	27
Trip5 2025	19	13/10/2025	08:15	19.8123 N, 48.2405 E	28.2	954	27
Trip5 2025	20	13/10/2025	09:04	19.8693 N, 48.4246 E	32.9	958	23
Trip5 2025	21	13/10/2025	09:30	19.9011 N, 48.4865 E	33	958	25
Trip5 2025	22	13/10/2025	10:10	19.9787 N, 48.6394 E	35.5	962	20
Trip5 2025	23	13/10/2025	10:56	20.0493 N, 48.7733 E	36.1	960	21
Trip5 2025	24	13/10/2025	11:39	20.0927 N, 48.8994 E	36.3	958	18
Trip5 2025	25	13/10/2025	11:55	20.1003 N, 48.9542 E	35	960	18
Trip5 2025	26	13/10/2025	12:16	20.1488 N, 49.0787 E	34.3	960	19
Trip5 2025	27	13/10/2025	12:42	20.2039 N, 49.1960 E	36.4	960	16
Trip5 2025	28	13/10/2025	13:08	20.2438 N, 49.3709 E	38.8	960	15
Trip5 2025	29	13/10/2025	13:56	20.2598 N, 49.5675 E	37.5	966	14
Trip5 2025	30	13/10/2025	14:13	20.2693 N, 49.6837 E	37.5	966	14
Trip5 2025	31	13/10/2025	14:46	20.2946 N, 49.8914 E	35.9	968	17
Trip5 2025	32	13/10/2025	15:38	20.3107 N, 50.0930 E	39	970	12
Trip5 2025	33	13/10/2025	16:10	20.3175 N, 50.3031 E	37.9	972	11
Trip5 2025	34	13/10/2025	16:58	20.3444 N, 50.5154 E	34.3	974	17
Trip5 2025	35	14/10/2025	06:58	20.3718 N, 50.7300 E	35.1	982	27

Expedition	Site	Date	Time	Coordinates	Temp (°C)	Press. (mbar)
Trip5 2025	36	14/10/2025	08:36	20.4232 N, 51.1448 E	33.2	982
Trip5 2025	37	14/10/2025	09:01	20.4542 N, 51.3449 E	32.3	984
Trip5 2025	38	14/10/2025	09:13	20.4610 N, 51.3702 E	33.2	984
Trip5 2025	39	14/10/2025	09:50	20.5102 N, 51.5974 E	37.2	986
Trip5 2025	40	14/10/2025	10:40	20.5472 N, 51.8256 E	39	984
Trip5 2025	41	14/10/2025	11:04	20.5398 N, 51.9461 E	37.1	986
Trip5 2025	42	14/10/2025	11:40	20.5552 N, 52.1240 E	36	984
Trip5 2025	43	14/10/2025	12:07	20.5676 N, 52.2241 E	40.4	984
Trip5 2025	44	14/10/2025	13:03	20.5678 N, 52.4104 E	37.1	984
Trip5 2025	45	14/10/2025	13:16	20.5635 N, 52.4678 E	39.5	984
Trip5 2025	46	14/10/2025	14:02	20.5537 N, 52.6692 E	38.6	982
Trip5 2025	47	14/10/2025	14:41	20.5418 N, 52.8688 E	36.2	984
Trip5 2025	48	14/10/2025	15:30	20.5686 N, 53.0940 E	36.7	986
Trip5 2025	49	14/10/2025	16:20	20.6447 N, 53.2879 E	35.2	986
Trip5 2025	50	14/10/2025	17:00	20.7139 N, 53.4123 E	34.7	984
Trip5 2025	51	15/10/2025	06:41	20.7987 N, 53.4870 E	31.1	990
Trip5 2025	52	15/10/2025	07:09	20.8278 N, 53.5784 E	33.5	990
Trip5 2025	53	15/10/2025	07:59	20.8515 N, 53.7579 E	33.7	990
Trip5 2025	54	15/10/2025	09:14	20.9202 N, 53.9622 E	37.5	992
Trip5 2025	55	15/10/2025	09:59	20.9023 N, 54.0024 E	39.3	988
Trip5 2025	56	15/10/2025	10:38	20.8216 N, 54.2239 E	42.8	988
Trip5 2025	57	15/10/2025	11:18	20.7350 N, 54.4176 E	39.4	986
Trip5 2025	58	15/10/2025	11:41	20.6654 N, 54.5933 E	41.8	986
Trip5 2025	59	15/10/2025	11:59	20.6563 N, 54.6359 E	40.2	984
Trip5 2025	60	15/10/2025	13:55	20.7064 N, 54.7321 E	35.9	988
Trip5 2025	oil spill	12/10/2025	10:20	19.4054 N, 45.7271 E	34.1	924
Trip5 2025	hot spring	14/10/2025	07:58	20.3972 N, 50.9655 E	32.7	984

Table 2: Mapping of XRF measurement qualities to ChEBI identifiers.

Analyte Quality	ChEBI Class
Ag concentration	http://purl.obolibrary.org/obo/CHEBI_131422 (CHEBI:131422)
Al concentration	http://purl.obolibrary.org/obo/CHEBI_26676 (CHEBI:26676)
Al2O3 concentration	http://purl.obolibrary.org/obo/CHEBI_28984 (CHEBI:28984)
As concentration	http://purl.obolibrary.org/obo/CHEBI_30188 (CHEBI:30188)
Au concentration	http://purl.obolibrary.org/obo/CHEBI_27563 (CHEBI:27563)
Ba concentration	http://purl.obolibrary.org/obo/CHEBI_29840 (CHEBI:29840)
BaO concentration	http://purl.obolibrary.org/obo/CHEBI_22694 (CHEBI:22694)
Bi concentration	http://purl.obolibrary.org/obo/CHEBI_131437 (CHEBI:131437)
Br concentration	http://purl.obolibrary.org/obo/CHEBI_22872 (CHEBI:22872)
Ca concentration	http://purl.obolibrary.org/obo/CHEBI_22927 (CHEBI:22927)
CaO concentration	http://purl.obolibrary.org/obo/CHEBI_22984 (CHEBI:22984)
Cd concentration	http://purl.obolibrary.org/obo/CHEBI_31344 (CHEBI:31344)
Ce concentration	http://purl.obolibrary.org/obo/CHEBI_22977 (CHEBI:22977)
CeO2 concentration	http://purl.obolibrary.org/obo/CHEBI_23081 (CHEBI:23081)
Cl concentration	http://purl.obolibrary.org/obo/CHEBI_131424 (CHEBI:131424)
Co concentration	http://purl.obolibrary.org/obo/CHEBI_23153 (CHEBI:23153)

Analyte Quality	ChEBI Class
CoO concentration	http://purl.obolibrary.org/obo/CHEBI_23356 (CHEBI:23356)
Cr concentration	http://purl.obolibrary.org/obo/CHEBI_50822 (CHEBI:50822)
Cr2O3 concentration	http://purl.obolibrary.org/obo/CHEBI_28073 (CHEBI:28073)
Cs concentration	http://purl.obolibrary.org/obo/CHEBI_50812 (CHEBI:50812)
Cu concentration	http://purl.obolibrary.org/obo/CHEBI_23053 (CHEBI:23053)
CuO concentration	http://purl.obolibrary.org/obo/CHEBI_28694 (CHEBI:28694)
Dy concentration	http://purl.obolibrary.org/obo/CHEBI_50823 (CHEBI:50823)
Er concentration	http://purl.obolibrary.org/obo/CHEBI_23883 (CHEBI:23883)
Eu concentration	http://purl.obolibrary.org/obo/CHEBI_23929 (CHEBI:23929)
Fe concentration	http://purl.obolibrary.org/obo/CHEBI_23992 (CHEBI:23992)
Fe2O3 concentration	http://purl.obolibrary.org/obo/CHEBI_18248 (CHEBI:18248)
Ga concentration	http://purl.obolibrary.org/obo/CHEBI_50819 (CHEBI:50819)
Gd concentration	http://purl.obolibrary.org/obo/CHEBI_27531 (CHEBI:27531)
Ge concentration	http://purl.obolibrary.org/obo/CHEBI_24185 (CHEBI:24185)
Hf concentration	http://purl.obolibrary.org/obo/CHEBI_27521 (CHEBI:27521)
Hg concentration	http://purl.obolibrary.org/obo/CHEBI_24467 (CHEBI:24467)
Ho concentration	http://purl.obolibrary.org/obo/CHEBI_25202 (CHEBI:25202)
I concentration	http://purl.obolibrary.org/obo/CHEBI_24610 (CHEBI:24610)
Ir concentration	http://purl.obolibrary.org/obo/CHEBI_24859 (CHEBI:24859)
K concentration	http://purl.obolibrary.org/obo/CHEBI_24873 (CHEBI:24873)
K2O concentration	http://purl.obolibrary.org/obo/CHEBI_26216 (CHEBI:26216)
La concentration	http://purl.obolibrary.org/obo/CHEBI_131423 (CHEBI:131423)
La2O3 concentration	http://purl.obolibrary.org/obo/CHEBI_24963 (CHEBI:24963)
Lu concentration	http://purl.obolibrary.org/obo/CHEBI_131424 (CHEBI:131424)
Mg concentration	http://purl.obolibrary.org/obo/CHEBI_25112 (CHEBI:25112)
MgO concentration	http://purl.obolibrary.org/obo/CHEBI_25107 (CHEBI:25107)
Mn concentration	http://purl.obolibrary.org/obo/CHEBI_31673 (CHEBI:31673)
MnO concentration	http://purl.obolibrary.org/obo/CHEBI_18291 (CHEBI:18291)
Mo concentration	http://purl.obolibrary.org/obo/CHEBI_131433 (CHEBI:131433)
Na concentration	http://purl.obolibrary.org/obo/CHEBI_28685 (CHEBI:28685)
Na2O concentration	http://purl.obolibrary.org/obo/CHEBI_26642 (CHEBI:26642)
Nb concentration	http://purl.obolibrary.org/obo/CHEBI_78659 (CHEBI:78659)
Nd concentration	http://purl.obolibrary.org/obo/CHEBI_25557 (CHEBI:25557)
Nd2O3 concentration	http://purl.obolibrary.org/obo/CHEBI_25490 (CHEBI:25490)
Ni concentration	http://purl.obolibrary.org/obo/CHEBI_131426 (CHEBI:131426)
NiO concentration	http://purl.obolibrary.org/obo/CHEBI_28112 (CHEBI:28112)
Os concentration	http://purl.obolibrary.org/obo/CHEBI_50824 (CHEBI:50824)
P concentration	http://purl.obolibrary.org/obo/CHEBI_25712 (CHEBI:25712)
P2O5 concentration	http://purl.obolibrary.org/obo/CHEBI_26004 (CHEBI:26004)
Pb concentration	http://purl.obolibrary.org/obo/CHEBI_37583 (CHEBI:37583)
PbO concentration	http://purl.obolibrary.org/obo/CHEBI_25016 (CHEBI:25016)
Pr concentration	http://purl.obolibrary.org/obo/CHEBI_131428 (CHEBI:131428)
PrO2 concentration	http://purl.obolibrary.org/obo/CHEBI_26257 (CHEBI:26257)

Analyte Quality	ChEBI Class
Pt concentration	http://purl.obolibrary.org/obo/CHEBI_131425 (CHEBI:131425)
Rb concentration	http://purl.obolibrary.org/obo/CHEBI_26156 (CHEBI:26156)
Re concentration	http://purl.obolibrary.org/obo/CHEBI_26421 (CHEBI:26421)
S concentration	http://purl.obolibrary.org/obo/CHEBI_26487 (CHEBI:26487)
SO3 concentration	http://purl.obolibrary.org/obo/CHEBI_26833 (CHEBI:26833)
Sb concentration	http://purl.obolibrary.org/obo/CHEBI_29324 (CHEBI:29324)
Sc concentration	http://purl.obolibrary.org/obo/CHEBI_22567 (CHEBI:22567)
Se concentration	http://purl.obolibrary.org/obo/CHEBI_26537 (CHEBI:26537)
Si concentration	http://purl.obolibrary.org/obo/CHEBI_27511 (CHEBI:27511)
SiO2 concentration	http://purl.obolibrary.org/obo/CHEBI_27573 (CHEBI:27573)
Sm concentration	http://purl.obolibrary.org/obo/CHEBI_30563 (CHEBI:30563)
Sm2O3 concentration	http://purl.obolibrary.org/obo/CHEBI_26523 (CHEBI:26523)
Sn concentration	http://purl.obolibrary.org/obo/CHEBI_131427 (CHEBI:131427)
Sr concentration	http://purl.obolibrary.org/obo/CHEBI_27051 (CHEBI:27051)
SrO concentration	http://purl.obolibrary.org/obo/CHEBI_26799 (CHEBI:26799)
Ta concentration	http://purl.obolibrary.org/obo/CHEBI_131435 (CHEBI:131435)
Tb concentration	http://purl.obolibrary.org/obo/CHEBI_26847 (CHEBI:26847)
Th concentration	http://purl.obolibrary.org/obo/CHEBI_26903 (CHEBI:26903)
Ti concentration	http://purl.obolibrary.org/obo/CHEBI_26986 (CHEBI:26986)
TiO2 concentration	http://purl.obolibrary.org/obo/CHEBI_27008 (CHEBI:27008)
Tm concentration	http://purl.obolibrary.org/obo/CHEBI_51050 (CHEBI:51050)
U concentration	http://purl.obolibrary.org/obo/CHEBI_26998 (CHEBI:26998)
V concentration	http://purl.obolibrary.org/obo/CHEBI_27197 (CHEBI:27197)
V2O5 concentration	http://purl.obolibrary.org/obo/CHEBI_27266 (CHEBI:27266)
W concentration	http://purl.obolibrary.org/obo/CHEBI_50825 (CHEBI:50825)
Y concentration	http://purl.obolibrary.org/obo/CHEBI_27250 (CHEBI:27250)
Yb concentration	http://purl.obolibrary.org/obo/CHEBI_27306 (CHEBI:27306)
Zn concentration	http://purl.obolibrary.org/obo/CHEBI_27303 (CHEBI:27303)
ZnO concentration	http://purl.obolibrary.org/obo/CHEBI_27363 (CHEBI:27363)
Zr concentration	http://purl.obolibrary.org/obo/CHEBI_36560 (CHEBI:36560)
ZrO2 concentration	http://purl.obolibrary.org/obo/CHEBI_27335 (CHEBI:27335)